DS	**Session**	Searching	**Question Information**	Level 1 Challenge 6

Problem Description:

Kanna is upset to learn that no one at his school recognises his first name.

Even his friends refer to him by his surname.

Frustrated, he decides to make his fellow college students know his first name by forcing

them to solve this question. The task is determining the third greatest number in the supplied array.

Constraints:

0<n<100

0<arr<1000

Input Format:

first line represents the number of elements N to be get

second line indicates input elements according to N

Output Format:

Single line represents the out put that is third largest number.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6
1 14 2 16 10 20
```

EXPECTED OUTPUT

```
The third Largest element is 14
```

Test Case 2

INPUT (STDIN)

```
7
19 -10 20 14 2 16 10
```

EXPECTED OUTPUT

```
The third Largest element is 16
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void thirdLargest(int
arr[],int arr_size)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
7
```

Test Case 2

TOKEN COUNT

```
220
```

Test Case 3

NLOC

```
44
```

Code

solution.c

```c
#include <stdio.h>
#include <limits.h>

void thirdLargest(int arr[], int arr_size) {
    if (arr_size < 3) return;
    int first = INT_MIN, second = INT_MIN, third = INT_MIN;
    for (int i = 0; i < arr_size; i++) {
        if (arr[i] > first) {
            third = second;
            second = first;
            first = arr[i];
        } else if (arr[i] > second && arr[i] != first) {
            third = second;
            second = arr[i];
        } else if (arr[i] > third && arr[i] != second && arr[i] != first) {
            third = arr[i];
        }
    }
    printf("The third Largest element is %d\n", third);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
    thirdLargest(arr, n);
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
6
1 14 2 16 10 20
The third Largest element is 14
PS C:\C Programming>
```